Modeling underwater sound propagation

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Underwater acoustic modeling must balance physical accuracy with computational cost for efficient applications of navigation, communication and environmental monitoring. This paper investigates this tradeoff by implementing two simple ray-based models (geometric ray tracing with Jacobian-based spreading and c-linear tracing with hybrid spherical-cylindrical spreading) and comparing them to the advanced Bellhop model. Both ray-based models aim to reproduce realistic sound propagation, transmission loss and bottom/surface reflections in a deep-water environment approximated with the Munk sound speed profile. Testing in a 100km deep-water scenario shows that both custom models successfully capture geometric ray structure, identifying 54 eigenrays compared to Bellhop's 24 energetically dominant paths. The Jacobian-based spreading model exhibits realistic amplitude variance reflecting caustic behavior, while the hybrid spreading model provides stable but less detailed amplitude predictions. Both approaches demonstrate computational efficiency while highlighting the tradeoff between geometric detail and physical accuracy in ray-based propagation modeling.

1 INTRODUCTION

Sound propagation in the ocean plays a central role in underwater positioning, navigation, timing and communication systems, which are applied in a wide range of fields from national security and exploration of marine resources to the study of the natural acoustic environment [1, 2]. However, accurately predicting underwater sound propagation remains a complex task due to environmental variability, ray bending, boundary interactions and different attenuation mechanisms.

To solve this, different computational models have been developed. Ray-based methods, that model the sound waves as discrete geometric paths are widely used due to computational efficiency. One particularly successful model that uses a variant of ray-tracing called gaussian beam tracing is the Bellhop model, which is a state-of-the-art model used in the field [3].

In this work, we created our own implementations of simple ray-based models and compared them to the advanced Bellhop model. In our implementations we tried two different approaches for both ray-tracing and spreading loss and evaluated their accuracy. In the comparison we focused on ray path accuracy, eigenray detection and impulse response at the receiver (i.e time and amplitude calculation of sound paths). The work was limited to a deep-water environment with long range (100km) between source and receiver.

2 BACKGROUND

2.1 Sound propagation in the ocean

In ocean acoustics, sound is propagated using the ocean as a waveguide, limited between the surface and the seafloor. Within this waveguide, speed of sound varies due to density, which in the ocean is related to static pressure, salinity and temperature. Many different empirical formulas for calculating the speed of sound have been proposed [1] with varying complexities and application ranges, but for most cases the simplified Medwin empirical formula can be used as a an approximation [4, 5]. In the Medwin formula, speed of sound (c, in m/s) is calculated as

$$\begin{aligned} c = {} & 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 \\ & + (1.34 - 0.01T)(S - 35) + 0.016z, \end{aligned} \tag{1}$$

where T is the temperature, in °C; S is the salinity, in parts per thousand, and z is the depth, in m.

Due to the variety of temperature, salinity and depth in the oceans around the world, underwater sound propagation is strictly dependent on the sound speed profile (SSP) that is used. A typical SSP in deep non-polar oceans has an unstable mixed layer near the surface and a stable, linearly increasing sound speed profile in the deep isothermal layer near the seafloor [4]. Between these regions exists the deep sound channel, where the speed of sound reaches a minimum. The Munk profile, is an idealized SSP showcased in Fig. 1, which contains many features of deep-water propagation and is often used for demonstrations of deep-water propagation.

For determining the sound propagation paths, speed of sound is treated in the same way the index of refraction is treated in optics. The refraction of the propagation direction changes as the speed of sound changes in accordance with Snell's law,

$$\frac{\cos\theta}{c} = p, \tag{2}$$

where p is a constant, θ is the ray angle and c is the local speed of sound [5]. Sound rays will therefore bend towards regions with lower speed of sound. A demonstration of deep-water propagation paths using the Munk profile is shown in Fig 1.

2.2 Reflections

At the surface and at the seabed of the ocean, the waves will experience reflections. In general, when the medium is not perfectly lossless, the local sound speed c becomes complex, and the reflection coefficient is therefore also complex. Each reflection introduces both an amplitude loss and a phase shift. The reflection coefficient $\mathscr{R}$ and transmission coefficient $\mathscr{T}$ at an interface between two fluid media can be expressed in terms of their specific acoustic impedances $\mathscr{Z}_i = \rho_i c_i / \sin\theta_i$ as

$$\mathscr{R} = \frac{\mathscr{Z}_2 - \mathscr{Z}_1}{\mathscr{Z}_2 + \mathscr{Z}_1}, \qquad \mathscr{T} = \frac{2\mathscr{Z}_2}{\mathscr{Z}_2 + \mathscr{Z}_1}. \tag{3}$$

When the two media have similar acoustic impedance ($\mathscr{Z}_1 \approx \mathscr{Z}_2$), the reflection coefficient is almost zero. Conversely, when medium 2 has a much larger impedance than medium 1, the reflection coefficient is close to 1, meaning that most of the energy is reflected.

2.2.1 Surface reflections

At the surface, the difference in acoustic impedance between air and water is so large that almost total reflection occurs ($\mathscr{R} \approx 1$). The surface can therefore be approximated as a perfectly reflecting boundary. However, this assumption breaks down in agitated conditions when wind and waves generate bubbles that scatter and absorb acoustic energy, leading to non negligible reflection losses [4].

In deep water, this bending traps the acoustic energy, hence creating ray paths that travel long distances without ever touching the bottom and that only reflect from the surface. Such typical deep-water environments, where these effects are prominent, occur in all oceans at depths exceeding about 2000m [4]. In this case, sound propagation often occurs without significant bottom interaction, dominated instead by refraction and surface reflection.

2.2.2 Bottom reflections

Reflection from the ocean bottom is far more complex. Unlike the nearly perfect surface, the bottom is generally a lossy boundary composed of sediments and rock layers with variable density, sound speed, and attenuation. The impedance mismatch between water and seabed materials determines how much energy is reflected versus transmitted into the sediment.

Layered sediments further complicate this behavior, producing frequency- and angle-dependent reflection losses due to interference within the layers. Depending on the incident angle, thin or half-wavelength layers can be acoustically transparent, while quarter-wavelength layers can act as anti-reflecting coatings [4].

2.3 Transmission loss due to seawater

Transmission loss describes the decrease in signal amplitude as it travels from a source to the receiver [4]. As sound waves travel in the ocean, their amplitude is progressively attenuated by various loss mechanisms contributing to the total transmission loss. While reflection loss and various scattering losses contribute to the total transmission loss, this section describes transmission loss due to propagation in the medium. The transmission loss due to the seawater can be divided into two parts: geometrical spreading loss and frequency-dependent absorption loss.

Geometrical spreading loss measures attenuation due to the signal spreading outwards from the source, assuming the medium to be lossless [4]. This spreading can in near field be assumed to be spherical, and is calculated as

$$TL_{spherical} = 20\log r, \tag{4}$$

where r is the horizontal range in meters. Due to the boundaries of the ocean surface and seafloor, the attenuation in far field can be expressed as cylindrical spreading loss

$$TL_{cylindrical} = 10\log r. \tag{5}$$

For a point source, we can make the near field assumption when the distance is less than the depth ($r < z_{max}$) followed by a transition region after which cylindrical spreading is applicable at longer ranges ($r \gg z_{max}$) [4].

In contrast, absorption loss measures how much of the acoustic energy is absorbed by the medium and turned into heat. This is usually modeled by an exponential decay law with distance x:

$$A = A_0 \exp(-\alpha x), \tag{6}$$

where A_0 is the rms amplitude at $x = 0$ and α is the attenuation factor. The coefficient α has some dependence on frequency, temperature, pressure, salinity, and acidity. However, a simplified expression depending only on frequency is considered sufficiently accurate for most problems in ocean acoustics [4, 6]:

$$\begin{aligned}\alpha = \frac{1}{8.69}\Big[3.3\cdot 10^{-3} + \frac{0.11 f^2}{1+f^2} + \frac{44 f^2}{4100+f^2} \\ + \quad 3\cdot 10^{-4} f^2\Big],\end{aligned} \tag{7}$$

where f is expressed in kHz and α in dB/km. Eq. 7 shows attenuation increases with frequency, such that low-frequency waves are only mildly attenuated, making them highly effective for long-range underwater communication compared to electromagnetic waves. For example, a 100Hz acoustic signal would need to travel around 2200 km to be attenuated by -10dB.

2.4 Volume scattering

In addition to boundary interactions at the sea surface and seafloor, sound propagation in the ocean is affected by volume scattering from marine life within the water column. Volume scattering is characterized by a scattering cross section, defined as the ratio of scattered power to incident intensity on a unit volume. This scattering contributes to both attenuation of coherent acoustic signals and generation of reverberation that interferes with sonar detection.

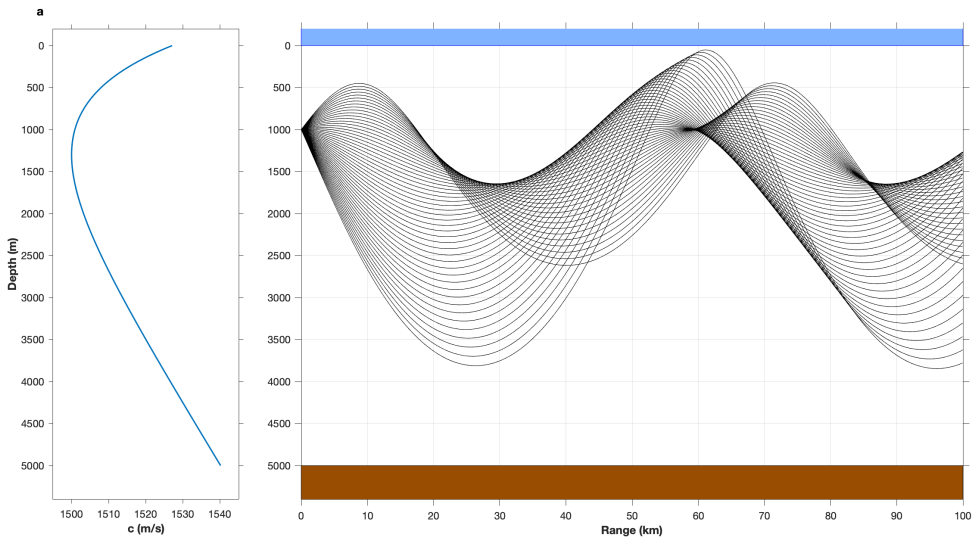

Fig. 1. Munk SSP (left) and sound propagation rays in Munk SSP (right). Ray tracing using c-linear method.

At lower frequencies (below 10 kHz), fish with gas-filled swim bladders are the dominant volume scatterers [4]. The swim bladder presents a high acoustic impedance contrast with the surrounding tissue and seawater, causing strong backscattering. Above 20 kHz, smaller organisms such as zooplankton become the primary scatterers. The target strength of individual fish depends on fish size, swim bladder morphology, frequency, and orientation relative to the acoustic beam, with typical values ranging from -40 to -20 dB for individual fish at moderate frequencies [7]. The swim bladder contribution accounts for 90-95% of the backscattered energy.

Volume scattering strength generally decreases with depth, with a notable exception being the deep scattering layer (DSL). The DSL is a concentrated layer of marine organisms, primarily small fish and zooplankton, that migrate vertically in the water column following the diurnal cycle. This layer can produce scattering returns comparable in magnitude to those from physical boundaries [4].

In acoustic propagation modeling, volume scattering introduces two main effects: attenuation of the coherent field through scattering loss, and generation of a diffuse reverberation field. The volume scattering strength S_v is expressed in decibels and relates the scattered intensity to incident intensity per unit volume. For a layer of thickness H, the integrated column scattering strength S_c can be computed by integrating the local volume scattering strength $s_v(z)$ over depth. These parameters are used in sonar performance prediction and acoustic propagation models for applications such as fisheries acoustics and marine mammal protection.

2.5 Sound propagation models

Underwater sound propagation can be described using several types of models, each based on different approximations of the acoustic wave equation. These models vary in their physical assumptions, numerical complexity, and suitability for particular frequency ranges or environmental conditions. This section briefly presents the most popular families of methods [4, 8].

First, ray theory based models describe sound as energy traveling along discrete geometric paths, or rays. This approach is particularly effective when the acoustic wavelength is small compared to the scale of environmental variations, such as at high frequencies or over long distances. The use of ray-based models in the research community was motivated by their simplicity and computational efficiency. However, they neglect diffraction and interference effects, making them less commonly used when it comes to model low frequency phenomena [9].

Next, the wave equation can be solved using the separation of variables technique to isolate the vertical and horizontal directions. The vertical wave equation solutions are normal modes, or eigenvalues, that can be summed at the source and receiver to generate the full acoustic field, depending on the horizontal propagation. Each mode behaves like a standing wave in depth and propagates horizontally with its own phase speed. This approach is particularly suited for low-frequency and long-range propagation, as it can accurately describe the interference patterns between modes. The main drawback of normal mode models is that they become too computationally expensive at high frequencies [8].

Finally, as most of the problems solved in ocean acoustics involve a one-way propagation (i.e. from source to receiver) separating the wave equation into incoming and outgoing solutions leads to the Parabolic Equation (PE) method. PE models can account for refraction, diffraction, and scattering, which explains why they are so widely used and popular in the research community [4]. The PE computational load increases with the square of the frequency, which generally limits its use to frequencies less than 1 kHz [8].

In summary, each category of sound propagation model offers a different balance between accuracy, complexity

and computational cost. The choice of the model is determined by the problem requirements.

3 METHODS

Three different underwater sound propagation models are compared in this work. The models *Geometric ray tracing* and *c-linear* are simple ray-based methods based on techniques represented in [4]. The *Bellhop* model is a more advanced beam tracing program using open source code from [3]. In the models geometric ray tracing and c-linear, different algorithms for calculating spreading loss were implemented.

3.1 Ray-tracing implementation

For both the implementations of geometric ray tracing and c-linear, ray paths are computed by discretizing the arclength s of the ray into small steps of size Δs and updating the horizontal and vertical positions as

$$x_{n+1} = x_n + \Delta s \cos\theta_n, \tag{8}$$

$$z_{n+1} = z_n + \Delta s \sin\theta_n, \tag{9}$$

where θ_n is evaluated at the current depth using Snell's law (Eq. 2). The difference in the two implementations: geometric ray tracing and c-linear, is how the ray-tracing is calculated within a step.

The environment is represented by a one-dimensional sound speed profile $c(z)$ that varies with depth z. A simple Munk-like profile [3] is employed to approximate realistic ocean conditions:

$$c(z) = c_0\left[1 + \varepsilon\left(\bar{z} - 1 + e^{-\bar{z}}\right)\right], \quad \bar{z} = \frac{2(z - z_0)}{z_0}. \tag{10}$$

where $c_0 = 1500$ m/s is the reference sound speed, $z_0 = 1300$ m is the reference depth (sound channel axis), and $\varepsilon = 0.00737$ is a non-dimensional perturbation parameter controlling the strength of the sound speed variation.

After implementing a source and receiver configuration, calculation of eigenrays is done using a ray fan. Ray-tracing begins from a point source, producing a certain amount of rays at even intervals between two angles, i.e a ray fan. Each individual ray is traced until the x-coordinate is the same as the receivers. Eigenrays are detected if the terminal depth of the ray is close to the depth of the receiver, allowing a tolerance of 5m.

3.1.1 Geometric Ray Tracing

The underwater acoustic propagation is first modeled using geometric ray tracing, governed by Snell's law. This simple method does not solve the full acoustic wave equation, but approximates the trajectories of sound waves as they travel through the horizontally stratified water environment. The trajectory of the ray is governed by the differential equations:

$$\frac{dx}{ds} = \cos\theta \qquad \frac{dz}{ds} = \sin\theta, \tag{11}$$

where s is the arclength along the ray. The local angle θ can be recovered from Snell's law:

$$\theta(z) = \arccos(p\,c(z)), \tag{12}$$

provided that $|p\,c(z)| \leq 1$, where p is the Snell's law constant.

At each step, reflections from the sea surface and bottom boundaries were handled by reversing the vertical direction of propagation:

$$\sin\theta_{n+1} = -\sin\theta_n, \tag{13}$$

whenever the ray reached $z = 0$ (surface) or $z = z_{\max}$ (bottom).

3.1.2 Cell method: c-linear

Cell methods refer to the practice of calculating rays by dividing the medium into smaller discretized elements with simple properties. This can be done for range-dependent mediums by dividing the medium into triangles [4]. A simple implementation of the cell method c-linear in a range-independent medium is presented.

Assuming linear sound speed within a discretized step, the c-linear method can be used for calculating ray tracing [4]. Compared to the method of geometric ray tracing, which interpolates the ray at a discretized step as a straight line from start to end, c-linear calculates the step as a circular arc based on the linear variation in depth g from the sound speed

$$c(z) = c_0 + gz, \tag{14}$$

where c_0 is the initial sound speed at the start of a step and z is the depth.

From the local angle in Snell's law derived in Eq. 12, the change in angle over the ray path can further be derived as

$$\frac{d\theta}{ds} = -\frac{p}{\sin\theta}\frac{dc}{ds} = -\frac{p}{\sin\theta}\frac{\delta c}{\delta z}\frac{dz}{ds}. \tag{15}$$

Since it is assumed that the change of speed of sound over depth is constant, and using relations in Eq. 11, the last two terms can be substituted as

$$\frac{d\theta}{ds} = -\frac{p}{\sin\theta} g \sin\theta = -gp, \tag{16}$$

which is a constant. A constant ray curvature $d\theta/ds$ signifies that the ray follows the path of a circular arc. The angle of the ray at the end of the step is

$$\theta_{n+1} = \theta_n + \frac{d\theta}{ds} ds = \theta_n - gp \cdot ds, \tag{17}$$

where ds is the step length and n is an arbitrary discretized step.

To calculate reflections, a boundary hit point within the step is calculated using linear interpolation if $z \geq z_{max}$ or $z \leq z_{min}$ and specular reflection is assumed as in Eq. 13.

3.2 Transmission loss implementation

In the implementation of transmission loss, three loss mechanisms where taken into account: geometrical spreading, absorption loss and reflection loss. Two different geometrical-spreading models were used in this work,

and implemented into one of the ray-tracing models. The implementations are however not dependent on the ray-tracing algorithm and could be implemented into either one. Absorption loss and reflection loss was implemented identically for both models.

For measuring the impulse response at the receiver, arrival time calculation is also needed. This is done within the ray-tracing algorithms for each step by multiplying the step length with the local speed of sound.

3.2.1 Geometrical spreading

(1) Hybrid Spherical-Cylindrical Spreading (c-linear Model)

In the c-linear ray tracer, geometrical spreading is modeled using a hybrid approach: spherical spreading in the near field and cylindrical spreading in the far field. The transition point is chosen as the depth of the ocean floor $R_{\text{tr}} = z_{max}$. The transmission loss due to purely geometrical divergence is expressed in decibels as

$$\text{TL}_{\text{geo}} = \begin{cases} 20\log_{10}(R), & R \le R_{\text{tr}}, \\ 20\log_{10}(R_{\text{tr}}) + 10\log_{10}(R - R_{\text{tr}}), & R > R_{\text{tr}}, \end{cases}$$

where R is the total acoustic path length. The corresponding amplitude factor is

$$A_{\text{geo}} = 10^{-\text{TL}_{\text{geo}}/20}. \tag{18}$$

This model provides the $1/R$ decay in the near field and the $1/\sqrt{R}$ decay associated with cylindrical spreading in deep-water waveguides. However, although computationally efficient, this hybrid model assumes an idealized waveguide geometry and therefore cannot represent changes in ray density caused by refraction. For these reasons the method provides only an approximate estimate of geometrical spreading, especially in environments with strong vertical sound-speed gradients (e.g. in the case of a Munk SSP).

(2) Jacobian-Based Geometrical Spreading (Geometric Ray Tracing Model)

In the geometric ray-tracing model, geometrical spreading is derived directly from ray theory using the Jacobian of the mapping between launch angle and spatial coordinates. Following [4], the ray amplitude at the receiver is

$$A_{\text{geo}} = \frac{1}{4\pi}\left|\frac{c(s)\cos\theta_0}{c(0)J(s)}\right|^{1/2}, \tag{19}$$

where θ_0 is the launch angle, $c(0)$ and $c(s)$ are the sound speeds at the source and receiver, and $J(s)$ is the geometric spreading factor

$$J(s) = \frac{r(s)}{\sin\theta(s)}\frac{\partial r(s)}{\partial\theta_0}. \tag{20}$$

Here, $r(s)$ is the horizontal range of the ray at arclength s, and $\partial r/\partial\theta_0$ is computed numerically by comparing the receiver crossing points of neighboring rays in the launch-angle fan. This formulation automatically captures shadow zones, focusing/defocusing regions, and the full structure of multipath propagation in a refracting ocean waveguide.

3.2.2 Absorption loss

Absorption loss was modeled as depending only on frequency of the signal, using Eq. 7 described in section 2.3. For each eigenray, the total geometric path length L (in meters) is converted to kilometers then affected by the attenuation factor α. Each ray was sent at a frequency $f = 100$ Hz.

3.2.3 Reflection loss

Reflection of acoustic rays at the ocean boundaries is implemented explicitly during the ray-tracing integration. The sea surface is modeled as a perfect reflection boundary. Therefore the pressure reflection coefficient is

$$R_{\text{surf}} = -1, \tag{21}$$

which introduces a phase shift of π (sign inversion).

The seafloor is treated as a fluid sediment layer characterized with density ρ_2 and sound speed c_2, while the water column has $\rho_1 = 1000$ kg/m^3 and sound speed $c_1 = c(z_{\max})$ [4]. The reflection coefficient for an incident angle θ_i is computed using the standard plane-wave fluid-fluid boundary formula:

$$R_{\text{bot}} = \frac{Z_2\cos\theta_i - Z_1\cos\theta_t}{Z_2\cos\theta_i + Z_1\cos\theta_t}, \quad Z_k = \rho_k c_k, \tag{22}$$

where the transmitted angle θ_t follows Snell's law.

3.3 Bellhop

Bellhop is a beam tracing program developed by Michael Porter as part of the Acoustics Toolbox [3]. Unlike simple ray-based methods that treat acoustic paths as infinitesimally thin lines, Bellhop employs Gaussian beam tracing, where each ray has a finite width. This approach can reduce artifacts near caustics and boundaries that are problematic for pure ray methods.

The Gaussian beam formulation extends traditional ray theory by associating a complex beam width parameter with each ray. As the beam propagates through the medium, both its center trajectory and width evolve according to coupled differential equations. For a beam with width parameter $q(s)$, the evolution along arclength s is governed by

$$\frac{dq}{ds} = \frac{c^2}{q}\nabla^2 c, \tag{23}$$

where c is the local speed of sound and $\nabla^2 c$ represents the second derivative of the sound speed profile. This formulation allows handling of regions where rays converge (caustics) without the singularities that occur in pure ray methods. At each receiver location, contributions from all beams are summed with appropriate amplitude and phase corrections based on beam geometry.

Bellhop's numerical integration uses an adaptive step size algorithm that automatically adjusts to ensure ray paths accurately land on discontinuities in the sound speed profile, such as at SSP points and boundary segments. This

approach ensures proper handling of refraction at layer interfaces and specular reflections at boundaries. The integrator follows the standard ray equations derived from Snell's law in stratified media, with additional terms to track beam spreading and curvature.

3.3.1 Configuration via .env Files

Bellhop reads its environmental parameters from text files with the '.env' extension, following a standardized format used throughout the Acoustics Toolbox. These files specify the complete acoustic environment including the sound speed profile, boundary conditions, source and receiver positions, and computational parameters.

The boundary specification allows three main types: Vacuum boundaries ('V') produce perfect reflection with no transmission; Rigid boundaries ('R') enforce zero normal particle velocity; and Acoustic halfspace boundaries ('A') model the seabed as a fluid halfspace with specified sound speed, density, and attenuation. For the acoustic halfspace option, parameters are given as depth, sound speed, shear speed (zero for fluid), density, compressional attenuation, and shear attenuation. These boundary models range from simple geometric reflection to frequency-dependent bottom interaction.

Bellhop supports multiple run types specified in the '.env' file: Ray tracing ('R') outputs the full geometric paths of all launched rays; Eigenray ('E') identifies only those rays connecting source to receiver; Coherent transmission loss ('C') computes the complex pressure field by summing beam contributions with phase information; Incoherent transmission loss ('I') sums intensities without phase; and Semicoherent ('S') uses a hybrid approach. The choice of run type will determine both the computational cost and the physical interpretation of results.

3.3.2 Coherent and Incoherent Summation

Bellhop can perform both coherent and incoherent field summation. In coherent mode ('C'), the complex acoustic pressure at each receiver is computed by summing contributions from all beams with their relative phases preserved:

$$p(\mathbf{r}) = \sum_{j=1}^{N} A_j e^{i\phi_j}, \tag{24}$$

where A_j and ϕ_j are the amplitude and phase of the j-th beam contribution, and N is the total number of beams reaching the receiver. This will produce interference patterns including constructive and destructive interference, which are physically realistic for coherent sources and receivers at fixed positions.

In incoherent mode ('I'), only intensities are summed, discarding phase information:

$$I(\mathbf{r}) = \sum_{j=1}^{N} |A_j|^2. \tag{25}$$

This approach is appropriate when modeling averaged fields over many source or receiver positions, or when the acoustic field has significant temporal or spatial incoherence. Incoherent summation produces smoother transmission loss patterns without the fine interference structure of coherent calculations. The simple ray-based methods presented in earlier subsections implicitly perform an operation similar to incoherent summation by counting ray arrivals without tracking phase.

4 RESULTS

All three models were tested using identical environmental parameters: a Munk sound speed profile with source at 1000m depth and receiver at 100km range and 1000m depth. A fan of 10,001 rays was launched between $-30°$ and $30°$, and the seabed was modeled as a sandy bottom with density $\rho = 1800$ kg/m^3 and sound speed $c = 1700$ m/s at 100 Hz.

4.1 Eigenray Detection

For this specific scenario (100km range, 1000m source and receiver depths, Munk profile), the three models found significantly different numbers of eigenrays connecting the source to the receiver. Both the c-linear curvature model and the geometric ray tracing model detected 54 eigenrays, while Bellhop identified 24. This 2-2.3x difference reflects distinct detection approaches rather than modeling accuracy or an inherent property of the models. The custom ray-based methods scan through a dense ray fan and flag any rays passing within a tolerance of the receiver depth. Bellhop uses a beam tracing approach with adaptive step sizing and beam convergence criteria, identifying only the energetically dominant paths. Different source-receiver geometries, ranges, or sound speed profiles would yield different eigenray counts for all three models.

Fig. 2 shows the eigenray paths for the c-linear curvature model, while Fig. 3 shows the geometric ray tracing model results. Both models capture the characteristic deep-water propagation pattern with rays refracting through the sound channel. The non-bouncing eigenrays launched near horizontal angles ($\theta_0 \approx \pm 4 - 5°$) remain trapped in the sound channel, while steeper launch angles produce rays that reflect from the surface and bottom.

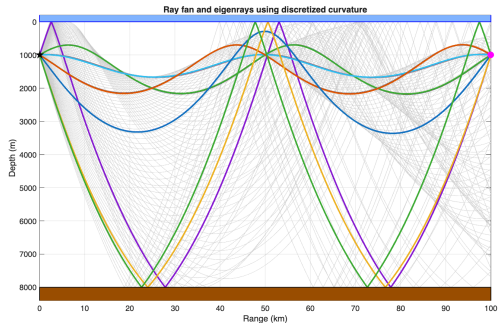

Fig. 2. Eigenray paths computed by the c-linear curvature model. Source at 1000m depth, receiver at 100km range and 1000m depth.

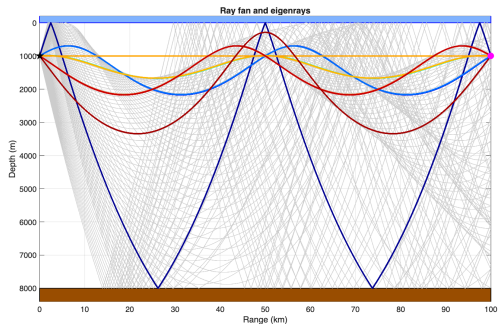

Fig. 3. Eigenray paths computed by the geometric ray tracing model. Same configuration as Fig. 2.

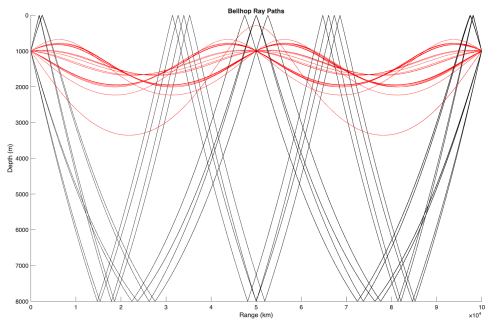

Fig. 4. Eigenray paths computed by Bellhop. Bellhop identifies fewer but energetically dominant paths compared to the custom models.

4.2 Ray Path Characteristics

The non-bouncing eigenrays cluster into three groups based on launch angle. Rays launched near $\theta_0 \approx \pm 4.7^\circ$ have path lengths around 100,215m and travel times of 66.66s. A second cluster near $\theta_0 \approx \pm 0.6 - 0.7^\circ$ shows slightly shorter paths (100,048m) with travel times of 66.664s. Rays launched at steeper angles ($\theta_0 \approx \pm 9.8^\circ$) travel longer paths (100,737m) with travel times of 66.58s. The c-linear model also detected several bouncing rays with 2 bottom and 2-3 surface reflections, reaching path lengths up to 105km and travel times approaching 68s. The geometric ray tracing model identifies an faulty eigenray near $\theta_0 = 0^\circ$ going directly horizontally to the receiver.

4.3 Amplitude Predictions

Amplitude calculations differ significantly between the two custom models due to their spreading loss approaches. The c-linear model uses hybrid spherical-cylindrical spreading, producing highly consistent amplitudes clustered around -128.15 to -128.16 dB for all non-bouncing rays. Bouncing rays with bottom and surface reflections are attenuated to approximately -148 dB due to reflection losses.

The geometric ray tracing model implements Jensen's Jacobian spreading formula, which tracks changes in ray tube cross-section along the propagation path. This produces highly variable amplitude predictions: non-bouncing rays range from -116 dB to -167 dB, showing the effect of caustic formation and ray convergence where ray tubes expand and contract. Travel times for the dominant arrivals cluster tightly around 66.66s, consistent with a 100km range and average sound speed near 1500 m/s.

Fig. 5 shows the arrival structure for the geometric ray tracing model (with Jacobian-based spreading) and Fig. 6 shows the arrival structure for the c-linear tracing model (with hybrid spherical-cylindrical spreading).

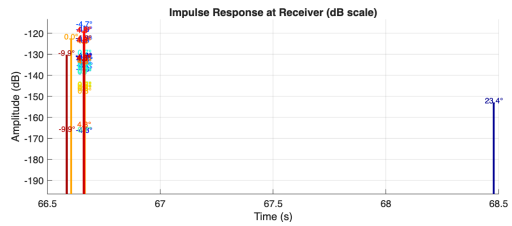

Fig. 5. Impulse response at receiver showing arrival times and amplitudes for the geometric ray tracing model.

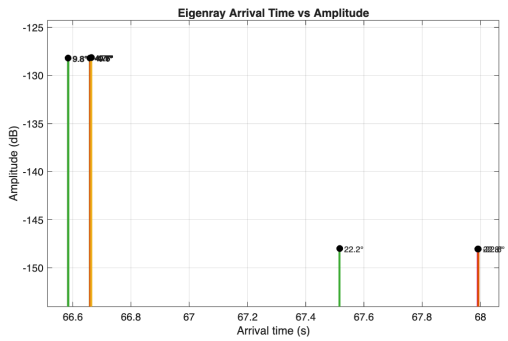

Fig. 6. Impulse response at receiver showing arrival times and amplitudes for the c-linear model.

5 DISCUSSION

The comparison reveals that simple ray-based methods can capture the geometric structure of deep-water propagation but differ in how they handle amplitude predictions and eigenray selection. The c-linear method improves on basic geometric ray tracing by treating each integration step as a circular arc rather than a straight line. This accounts for the continuous curvature of rays in a stratified medium and produces smoother trajectories, particularly in regions with strong sound speed gradients.

The geometric ray tracing method further refines amplitude calculation by tracking the Jacobian determinant along the ray path. This captures focusing and defocusing effects as ray tubes expand and contract. The large amplitude variance in the ray parameter model (-116 to -167 dB for non-bouncing rays) reflects realistic caustic behavior where rays converge and diverge. However, the method also shows some outliers with very high attenuation (-167 dB), which may indicate numerical issues near turning points where rays approach horizontal or at caustics where the Jacobian becomes singular.

Bellhop identifies fewer eigenrays because it selects beams based on energy contribution rather than geometric proximity. The 24 eigenrays Bellhop found likely represent the dominant propagation paths that carry most of the acoustic energy. The custom models, by scanning through 10,001 rays with a fixed tolerance criterion, capture many weakly excited paths that contribute minimally to the total field. This distinction is important for applications: if only the arrival structure is needed, the custom models provide more detail. If accurate pressure field prediction is required, Bellhop's beam summation with proper phase handling would be more appropriate.

One limitation of the custom models is their treatment of boundary reflections. Both assume perfect specular reflection with phase reversal at the surface, which is realistic for the air-water interface. At the bottom, both models compute angle-dependent reflection coefficients using the fluid-fluid boundary formula with the incident angle from Snell's law. Real sediment layers introduce additional frequency-dependent losses and phase shifts from layered structure that are not captured by the simple halfspace model used here.

Both custom models include frequency-dependent absorption loss. However, at a signal frequency of 100 Hz, seawater absorption contributes negligibly to transmission loss over 100km ($\sim$0.5 dB total), but at higher frequencies this would become significant. Both models neglect volume scattering from marine life and small-scale fluctuations in the sound speed profile, which would introduce additional spreading and loss mechanisms.

The computational performance difference between the methods is substantial. The geometric ray tracing method completes in under a second due to its simple integration steps. The c-linear method takes slightly longer because it computes circular arc parameters at each step. However, the geometric ray tracing method, despite optimizations like pre-allocated arrays and receiver-only data tracking, runs slower due to the Jacobian calculation. Bellhop's adaptive step sizing and beam tracking make it more expensive per ray, but it launches fewer beams to achieve similar coverage.

6 CONCLUSION

We implemented two simple ray-based underwater sound propagation models and compared them to the established Bellhop beam tracer. The c-linear curvature method and ray parameter method both successfully reproduce the geometric structure of long-range deep-water propagation through a Munk sound speed profile. In the tested scenario (100km range, 1000m source and receiver depths), both custom models identified 54 eigenrays compared to 24 found by Bellhop.

The difference in eigenray count reflects detection philosophy rather than accuracy, and these numbers are specific to this scenario. The custom models scan densely spaced rays and report all paths passing near the receiver, while Bellhop selects energetically dominant beams. For impulse response prediction, the custom methods provide detailed arrival structure but require careful interpretation of amplitude predictions. The geometric ray tracing model's Jacobian spreading calculation captures realistic focusing effects but shows numerical sensitivity near caustics. The c-linear model's simpler geometric spreading produces stable results but misses fine-scale amplitude variations.

Both custom models include frequency-dependent absorption loss but neglect volume scattering. The boundary conditions use angle-dependent reflection coefficients but assume simplified halfspace sediment models. These approximations are acceptable for the 100 Hz, 100km scenario tested but would require refinement at higher frequencies or in shallow water environments. For accurate pressure field prediction including interference effects, methods like Bellhop that handle coherent beam summation remain necessary.

The work demonstrates that ray-based methods provide computational efficiency and physical insight for long-range propagation problems where geometric effects dominate over wave phenomena. The custom propagation models, used parameters and instructions on how to use the Bellhop model can be found in the following GitHub repository: `https://github.com/otahontas/ocean_acoustics_code`.

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